

Sudip Roy

Systems Software Engineer

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Summary

Systems-focused software engineer with hands-on experience building performance-sensitive Rust, Python, and TypeScript systems. Built and operated a multi-service platform used by a 50,000+ member community, with production work spanning low-latency IPC, storage engine tradeoffs, graph-based algorithm research, and performance-driven back-end design. Interested in compiler-adjacent, infrastructure, and runtime-heavy engineering problems.

Technical Skills

Languages: Rust, Python, TypeScript, SQL, JavaScript

Systems & Runtime: Linux, gRPC, UNIX domain sockets, concurrency, memory safety, low-latency service design

Algorithms & Performance: bitmasks, graph construction, constraint propagation, benchmarking, profiling, caching

Frameworks & Tools: Axum, Actix, Docker, Git, GitHub Actions, Redis, PostgreSQL, SQLite, libSQL, MongoDB

Experience

Software Developer

2020 – Present

Worldwide Discord Platform

Rust, Python, TypeScript

- Built and maintained a production multi-service platform serving 50,000+ members and 2,000+ daily active users across Python, Rust, and TypeScript services.
- Designed a three-layer architecture with Python handling the event gateway, Rust owning the performance-critical data and transaction path, and Next.js supporting admin workflows.
- Moved critical backend logic into Rust because GC pauses were unacceptable on the financial path, improving predictability for latency-sensitive operations.
- Introduced gRPC boundaries between services and optimized same-machine communication using UNIX domain sockets, reducing local IPC latency by roughly 60–70%.
- Evolved storage from raw JSON to MongoDB, Redis, PostgreSQL, and embedded libSQL based on concrete concurrency, integrity, and latency failures in production.
- Built Rust-side caching and embedded libSQL execution paths to remove avoidable network round-trips and keep hot operations close to the process.

Projects

Sudoku Solver Research

Active

github.com/drlove2002/sudoku_solver

Rust, Criterion, Algorithm Research

- Building a modular Sudoku-solving and benchmarking system in Rust to evaluate interchangeable solving components on a large real-world dataset of about 3 million puzzles.
- Using minigrid-based constraint propagation, permutation generation, compatibility graph construction, and graph export for analysis instead of a single hardcoded solver path.
- Treating the project as a research and benchmarking problem: compare solver combinations, measure behavior, and produce paper-quality technical documentation in LaTeX.

Pattern Pursuit

2023

github.com/drlove2002/pattern_pursuit

Rust, Actix, TypeScript, Redis

- Built an interactive prediction game using a Rust and Actix backend, TypeScript frontend, and Redis-backed real-time state.
- Applied 5-gram statistical language modeling to drive prediction logic and adaptive gameplay behavior.

Education

M.Sc. in Computer Science

Final Year

University of Calcutta

West Bengal, India

B.Sc. in Computer Science

2020 – 2023

University of Calcutta

West Bengal, India

Additional

Interests: systems programming, compilers, runtimes, distributed systems, performance engineering

Languages: Bengali (Native), English (Fluent), Hindi (Intermediate), German (A1)